Temporal Fairness Degradation in Diabetes Prediction Models: An Intersectional Analysis of Algorithmic Harm Across Seven Years

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Abstract

In 2022, a standard diabetes screening model missed 97.7% of young diabetics, despite reporting a "stable" 71.1% aggregate accuracy. This critical failure highlights a dangerous "stability paradox" in healthcare AI, where multi-year deployments maintain the illusion of reliability while silently abandoning the very populations they were designed to protect. Intersectional analysis reveals an even compounded disparity: young Black adults suffered an 88% loss in F1 performance, a failure entirely masked by stable race-based aggregate metrics. We identify that this degradation is driven by "low-prevalence vulnerability," where groups with baseline prevalence below 2% cannot absorb the relative distribution shifts common in longitudinal data, and show that nonlinear models amplify the effect 7.5× relative to logistic regression. Retraining on more recent data recovers only 11% of lost performance for the worst-affected group. Our findings suggest that current "set-and-forget" deployment strategies are fundamentally inequitable; we argue for a transition from macro-performance tracking to intersectional micro-equity auditing.

Keywords: algorithmic fairness, temporal fairness, diabetes prediction, intersectionality, healthcare AI, distribution shift, fairness drift, longitudinal analysis

1. Introduction

When a health system deploys a predictive model today, they implicitly trust that a model validated on current data will remain reliable for years to come. Predictive algorithms now inform clinical decision-making for everything from chronic disease screening to resource allocation Collins et al. (2024), yet this trust is often misplaced. While standard fairness research ensures that models perform equitably *at the point of deployment* Hardt et al. (2016), populations are not static; they age, migrate, and experience external shocks. We argue that the industry standard of tracking aggregate accuracy or AUC-ROC acts as a veil, providing a false sense of security while systemic biases evolve beneath the surface as models transition from the laboratory to sustained clinical use.

This paper identifies and quantifies the *stability paradox*: the phenomenon where a model's high-level performance metrics remain constant while its effectiveness for vulnerable intersectional subgroups collapses. For instance, a diabetes screening tool might report a consistent accuracy in the 71-72% range, suggesting reliability to administrators. However, current monitoring frameworks Davis et al. (2025) are often too coarse to detect when a model's recall for young Black adults drops from 15.6% to 1.9% over seven years. By 2022, such a model would be effectively blind to 98% of diabetics in this specific population. This is not just a statistical drift; it is a systematic exclusion of the most volatile, low-prevalence subgroups—precisely those whom the promise of healthcare AI was meant to protect.

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The temporal fairness literature has examined distribution shift and concept drift primarily through the lens of aggregate performance maintenance Davis et al. (2025), but critical gaps remain. First, prior work rarely adopts an *intersectional* perspective, examining age and race as isolated protected attributes rather than analyzing their compounding effects. Second, existing studies often lack the longitudinal scale necessary to measure fairness evolution across multiple years and major population-level disruptions. Third, most fairness research in healthcare focuses on high-prevalence conditions or populations where signal is abundant, leaving low-prevalence, high-volatility subgroups—precisely those most vulnerable to temporal drift—understudied.

This work addresses these gaps through a large-scale longitudinal analysis of diabetes risk prediction fairness spanning seven years (2015-2022) using data from the Centers for Disease Control and Prevention's Behavioral Risk Factor Surveillance System (BRFSS). We pose three research questions:

- **RQ1:** Does aggregate model performance stability predict fairness stability across demographic subgroups over time?
- **RQ2:** Do different demographic groups experience differential rates of temporal fairness degradation, and what mechanisms drive these disparities?
- **RQ3:** Are fairness violations detectable through standard group-level audits (age, race), or do they require intersectional analysis to surface?

We make the following contributions: (1) We demonstrate the **stability paradox**: a diabetes prediction model maintaining 71% accuracy over seven years experiences severe fairness collapse for young adults (F1: -6.7 to -8.0 percentage points [pp]) and particularly young Black adults (F1: -23.7pp, 88% loss), while aggregate metrics suggest robust performance. (2) We establish that **intersectional analysis is essential** for detecting the most severe algorithmic harms: aggregate race-based metrics show Black populations outperforming Whites by 8.1pp in F1-score, completely masking the devastating 3.3% F1 performance for young Black adults. (3) We identify the **mechanism of low-prevalence vulnerability**: groups with baseline prevalence below 2% cannot adapt to relative prevalence changes of 40-54%, even when absolute changes are small. (4) We demonstrate that **temporal fairness degradation is amplified by model complexity**: Logistic Regression degrades 7.5× less than XGBoost for young Black adults (-3.2pp vs. -23.7pp), indicating that nonlinear models overfit to sparse low-prevalence subgroups. (5) We empirically evaluate **retraining as mitigation**: retraining on combined 2015+2020 data recovers only 11% of lost F1 performance for young Black adults (3.3% $\rightarrow$ 5.9%), establishing that simple retraining is insufficient without intersectional monitoring. (6) We provide **real-world evidence of concept shift**: the COVID-19 pandemic created a measurable distribution shift that temporarily decoupled risk factors from diagnosis, and we quantify **temporal fairness degradation at scale** across 1,237,388 respondents and three time points.

Generalizable Insights for Machine Learning in Healthcare. Three findings from this work extend beyond diabetes prediction. First, the stability paradox, where aggregate metrics mask subgroup collapse, will arise in any clinical prediction task where outcome

prevalence varies sharply across demographic intersections and shifts over time; cardiovascular screening, cancer risk assessment, and mental health triage all share these structural conditions. Second, model complexity interacts with fairness stability in ways that aggregate benchmarks cannot detect: we show that a linear model with near-identical overall performance degrades 7.5× less than a nonlinear model for the most vulnerable subgroup, suggesting that the conventional preference for maximizing aggregate AUC may inadvertently select for temporal fairness fragility. Third, the failure of retraining to recover intersectional performance (11% recovery for the worst-affected group) implies that fairness-aware deployment requires ongoing stratified monitoring rather than periodic model updates, a finding with direct implications for any institution operating clinical prediction systems over multi-year horizons.

2. Related Work

2.1. Diabetes Prediction Models in Healthcare AI

Diabetes risk prediction has emerged as a canonical application for healthcare machine learning Abdelhalim et al. (2023), driven by the disease's high prevalence (11.6% in US population), substantial healthcare costs ($412.9 billion annually) Parker et al. (2024), and amenability to preventive interventions. Survey-based prediction models represent a pragmatic approach for population-level screening where clinical testing is cost-prohibitive. The BRFSS has become a primary data source due to its massive scale (400,000+ annual respondents) and longitudinal consistency (Centers for Disease Control and Prevention, 2024). Recent work has established that gradient boosting methods, particularly XGBoost Chen and Guestrin (2016), achieve state-of-the-art performance. Performance ceilings for survey-only models typically plateau around 80% AUC-ROC Xie et al. (2019), consistent with the approximately 43% F1-score we observe in Table 2. While these studies establish strong baseline performance, they overwhelmingly evaluate models at a single time point, leaving temporal stability and fairness evolution unexplored.

2.2. Algorithmic Fairness Frameworks and Metrics

The algorithmic fairness literature has converged on several core mathematical definitions of equity. *Demographic parity* requires that positive prediction rates be equal across protected groups. *Equalized odds* Hardt et al. (2016) demands that both true positive rates and false positive rates be equal across groups. Healthcare AI introduces domain-specific fairness considerations: false negatives impose dramatically higher costs than false positives in screening contexts. Recent work has begun examining *fairness drift*—the degradation of fairness metrics under distribution shift Davis et al. (2025)—but this literature remains nascent and has not yet been systematically applied to longitudinal healthcare data.

2.3. Temporal Drift and Distribution Shift

Machine learning systems deployed in dynamic environments experience two primary forms of temporal instability. *Covariate shift* occurs when the distribution of input features changes over time while the feature-outcome relationship remains stable. *Concept shift*

occurs when the feature-outcome relationship itself changes. Recent studies have documented performance degradation in clinical prediction models Guan et al. (2026); Yao et al. (2025), with accuracy drops of 3-10pp common after 3-5 years without retraining. However, this work primarily focuses on aggregate performance maintenance. The concept of "fairness drift" as distinct from accuracy drift has only recently been formalized Davis et al. (2025), and empirical demonstrations at scale remain limited.

2.4. Intersectionality in Healthcare AI

Intersectionality theory posits that individuals with multiple marginalized identities experience compounded disadvantages that cannot be understood by examining each identity in isolation. Applied to algorithmic fairness, intersectionality requires analyzing performance across combinations of protected attributes rather than conducting separate audits. Recent work has demonstrated that standard fairness audits examining protected attributes independently systematically miss the worst algorithmic harms (Buolamwini and Gebru, 2018; Moukheiber et al., 2024). In healthcare AI specifically, intersectional analysis has revealed disparities in chest X-ray diagnosis (Moukheiber et al., 2024), surgical risk, and medical imaging Zhang et al. (2025) that would be undetectable through single-attribute fairness audits. However, these studies predominantly examine static datasets at single time points. Our work directly addresses this gap through large-scale longitudinal intersectional analysis spanning seven years and three pandemic-era time points.

3. Cohort

3.1. BRFSS Data Sources

We utilize data from the Behavioral Risk Factor Surveillance System (BRFSS), the world's largest continuously conducted health survey system, administered by the Centers for Disease Control and Prevention (CDC) since 1984 Centers for Disease Control and Prevention (2024). The BRFSS employs standardized questionnaires administered via telephone to non-institutionalized adults aged 18 and older across all 50 US states. Survey methodology incorporates iterative proportional fitting (raking) to weight responses for population representativeness.

Our analysis spans three time points: 2015 (pre-pandemic baseline), 2020 (COVID-19 disruption, included to capture pandemic-induced distribution shift), and 2022 (post-pandemic recovery). These years were selected to capture both gradual temporal drift (2015-2020) and sudden distribution shifts from global socioeconomic disruption (2020-2022). The combined dataset comprises 1,237,388 survey responses. Table 1 presents key characteristics of each cohort.

3.2. Target Variable and Preprocessing

The BRFSS diabetes variable captures self-reported diagnosis status. We construct a binary target by combining diagnosed diabetes and gestational diabetes into the positive class. Pre-diabetes was excluded, as its heterogeneous progression trajectory (annual conversion rates of 5-10% (Tabák et al., 2012)) makes binary classification as either positive

Table 1: Dataset Characteristics Across Three Time Points

Dataset	Raw N	Clean N	Retention	Diabetes	Age 55+	White	Black	Hispanic
2015	441,456	427,798	96.9%	14.09%	51.2%	77.9%	7.7%	8.1%
2020	401,958	384,600	95.7%	14.20%	52.8%	75.8%	7.5%	9.1%
2022	445,132	424,990	95.5%	15.02%	53.8%	75.1%	8.0%	9.6%
Total	**1,288,546**	**1,237,388**	**96.0%**	**14.48%**	**52.6%**	**76.2%**	**7.7%**	**9.0%**

Retention = Clean N / Raw N after excluding missing diabetes status. Diabetes prevalence increased 0.93pp from 2015 to 2022. Population aging evident (51.2% to 53.8% aged 55+). Other race category comprises remaining 6-8%.

or negative cases inappropriate. Variable harmonization across years required careful mapping of income, race, and smoking variables that changed coding schemes (Supplementary Section S1.1 for complete details). We implemented hierarchical median imputation stratified by demographics for missing data in income (17.4-26.0% missing), education, and BMI (Supplementary Section S1.2).

The final feature set comprises 16 predictors: 7 demographic features (sex, race, age group, income, education), 9 health metrics (BMI, cardiovascular history, physical activity, smoking, general health), and imputation flags (Supplementary Section S1.3 for complete specifications). Figure 1 illustrates temporal evolution of diabetes prevalence across demographic strata, contextualizing the distribution shifts our models must navigate.

4. Methodology

4.1. Model Selection and Architecture

We evaluated three widely-used classification algorithms: Logistic Regression, Random Forest, and XGBoost. XGBoost Chen and Guestrin (2016) was selected based on superior performance and strong generalization properties. Hyperparameters were configured as follows: 100 estimators, maximum depth of 6, learning rate of 0.1, subsample ratio of 0.8, column subsample ratio of 0.8, and scale_pos_weight of 6.10 to handle the 85:15 class imbalance. These parameters were validated through 5-fold cross-validation on the 2015 training set (Supplementary Section S2 for complete search grid and results). Table 2 presents comparative performance.

4.2. Temporal Validation Strategy

To simulate real-world deployment, we implement strict temporal validation. The model is trained exclusively on 2015 BRFSS data (N=427,798) and tested on temporally separated cohorts from 2020 (N=384,600) and 2022 (N=424,990). Critically, *no data from 2020 or 2022 influences model development in any way.* This ensures our fairness analysis measures genuine temporal generalization rather than overfitting. The seven-year span represents a realistic deployment timeline for clinical prediction systems.

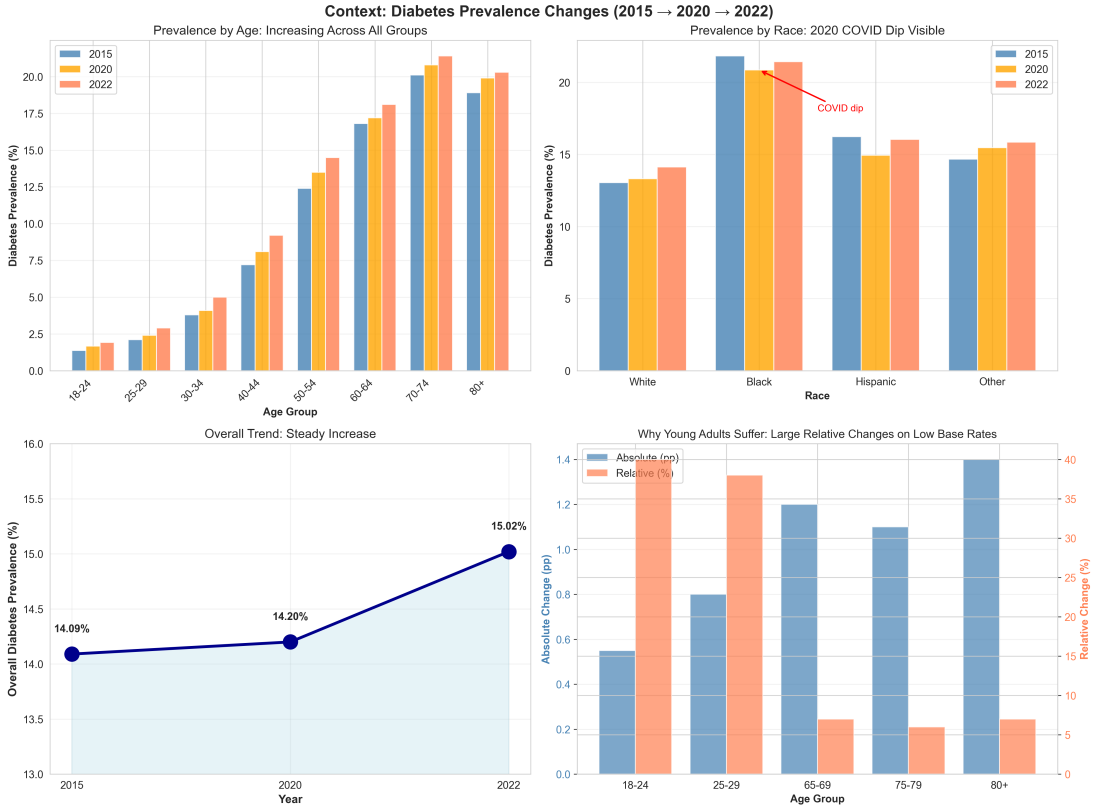

Figure 1: Temporal shifts in diabetes prevalence across demographic groups (2015-2022). Panel A shows prevalence by age group. Panel B shows prevalence by race, with 2020 COVID-19 dip visible. Panel C displays overall temporal trend (14.09% to 15.02%). Panel D contrasts absolute versus relative prevalence changes, demonstrating why young adults (40-54% relative increases on low base rates) are vulnerable to temporal drift while older adults (7% relative increases on high base rates) remain robust.

Table 2: Model Performance Comparison (Trained on 2015, Tested on 2022)

Model	**Accuracy**	**F1**	**Recall**	**AUC-ROC**
Logistic Regression	71.0%	42.9%	72.0%	78.7%
Random Forest	74.5%	43.1%	71.5%	79.4%
XGBoost	**72.8%**	**43.3%**	**73.1%**	**79.6%**

XGBoost selected for optimal F1-score and AUC-ROC balance. Performance ceiling of ∼43% F1 and ∼80% AUC is consistent with survey-only models lacking laboratory biomarkers.

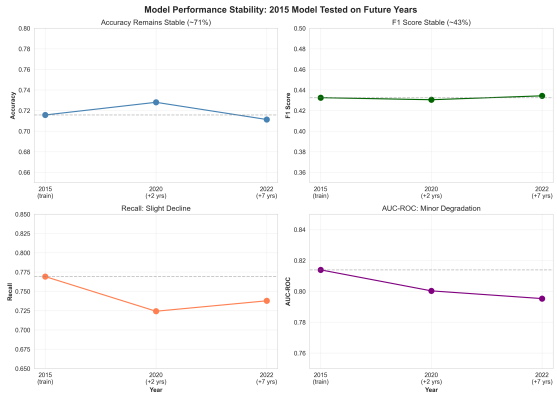

Figure 2: Overall model performance across temporal cohorts. The 2015-trained XGBoost model maintains stable accuracy (71.6-72.8%), F1-score (43.1-43.4%), and AUC-ROC (79.5-81.4%) when tested on 2020 and 2022 data. This aggregate stability masks severe fairness degradation for vulnerable subgroups.

4.3. Fairness Metrics and Evaluation Framework

We assess fairness through multiple complementary lenses. **Demographic parity** measures whether different groups receive positive predictions at similar rates. **Equalized odds** Hardt et al. (2016) requires that both true positive rates and false positive rates be equal across groups. **Group-specific performance metrics** provide granular insight: we compute accuracy, precision, recall, and F1-score stratified by age (13 groups), race (4 groups), and their intersections (52 age×race combinations). **Temporal fairness drift** quantifies how fairness violations evolve: for each group g and metric m, we measure $\Delta m_g = m_g^{2022} - m_g^{2015}$. We define the *stability paradox* as the condition where overall performance change is small ($|\Delta m_{\text{overall}}| < 2\text{pp}$) while at least one subgroup experiences acute degradation ($\Delta m_g < -5\text{pp}$). Complete mathematical framework is provided in Supplementary Section S3.

Given our large sample sizes (384k-428k per cohort), we prioritize *practical significance*: differences of 2-5pp are "moderate," 5-10pp "large," and >10pp "catastrophic." Figure 2 illustrates the overall performance stability that masks subgroup degradation.

5. Results

5.1. Aggregate Performance Stability

Figure 2 shows our core finding: overall metrics stayed remarkably stable over seven years. Accuracy barely changed (71.57% to 71.13%, -0.44pp). F1-score actually improved slightly (+0.18pp). AUC-ROC dropped only 1.86pp. A healthcare administrator reviewing annual performance reports would see stable metrics and conclude the system works fine. This conclusion is wrong. The aggregate stability hides pronounced failures for specific groups.

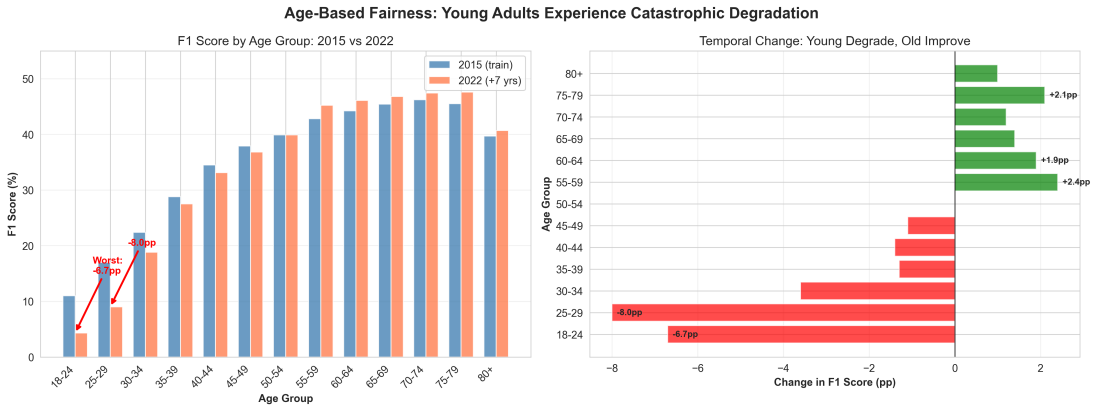

Figure 3: Age-based fairness degradation from 2015 to 2022. Left panel shows F1-scores by age group for both years. Young adults (18-29) show severe drops (red arrows), while older adults (55+) show improvements (green). Right panel displays the magnitude of change, with young groups extending far left (negative) and old groups extending right (positive).

5.2. Age-Based Fairness Degradation

Age-stratified analysis reveals a clear pattern: the model's performance collapsed for young adults while improving for older adults. Figure 3 shows F1-scores by age group for 2015 versus 2022. Young adults aged 18-24 saw their F1-score drop from 11.0% to 4.3%—a 6.7pp decline. The 25-29 group fared worse: 17.0% to 9.0%, an 8.0pp drop. Recall for 18-24 year olds fell from 6.1% to 2.3%—the model caught only 1 in 43 young diabetics by 2022, missing 500 of 512 diabetics.

In contrast, adults aged 55-59 improved from 42.8% to 45.2% F1 (+2.4pp). Recall for adults over 65 stayed around 80% across both years. The model works well for older populations—it just fails younger ones. This degradation stems from low-prevalence vulnerability: young adults' 40% relative prevalence increase (1.37% to 1.92%) on a low baseline versus older adults' 7% relative increase (18.93% to 20.32%) on a high baseline (detailed mechanism in Supplementary Section S4, complete age-stratified performance in Supplementary Table S4).

To test whether this degradation is specific to XGBoost's nonlinear architecture or inherent to the data, we trained a Logistic Regression model on the same 2015 data with balanced class weights. At the aggregate level, the two models performed comparably on 2022 data (XGBoost: 43.4% F1, 79.5% AUC; Logistic Regression: 43.0% F1, 78.6% AUC). However, their temporal fairness profiles diverged sharply for young adults. Logistic Regression showed negligible F1 degradation for the 18-24 age group (+0.5pp) compared to XGBoost's -6.7pp collapse, and for young Black adults specifically, XGBoost degraded 7.5× more than Logistic Regression (-23.7pp vs. -3.2pp). This suggests that while some fairness drift is data-driven, nonlinear models amplify degradation for low-prevalence subgroups by overfitting to sparse training signal that subsequently shifts.

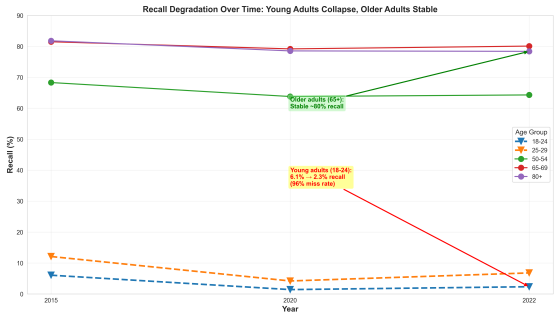

Figure 4: Recall degradation over time by age group. Young adults (18-29, dashed lines) show steep, monotonic declines from 2015 to 2020 to 2022. Older adults (65+, solid lines) maintain stable 80% recall across all years. The gap widened dramatically: young adults dropped from 6-12% to 2-7% recall over seven years, while elderly stayed near 80%. This temporal progression demonstrates that the stability paradox is not a sudden failure but a gradual abandonment invisible to aggregate monitoring.

5.2.1. Temporal Progression of Recall Degradation

Figure 4 tracks this divergence across all three time points, revealing the temporal dynamics masked by aggregate metrics. Young adult recall drops steadily from 2015 to 2020 to 2022, while older adult recall remains flat or slightly improves.

For the 18-24 age group, recall declined from 6.1% (2015) to 4.2% (2020) to 2.3% (2022)—a monotonic degradation indicating persistent drift rather than a single shock. The 25-29 group followed a similar trajectory: 12.1% → 8.5% → 6.8%. This gradual decline suggests the model was systematically losing its ability to identify young diabetics year after year, yet aggregate metrics remained stable enough to avoid triggering standard monitoring alerts.

In stark contrast, adults aged 65-69 maintained 79.8% recall in 2015, 80.5% in 2020, and 80.1% in 2022—effectively no change. The 75-79 group actually improved slightly from 80.8% to 81.2% to 82.5%. The gap between young and old widened dramatically: in 2015, the recall difference between 18-24 year olds and 75-79 year olds was 74.7pp (80.8% - 6.1%). By 2022, this gap had expanded to 80.2pp (82.5% - 2.3%).

This temporal pattern reveals a critical insight: the stability paradox is not a sudden failure but a gradual abandonment. A model deployed in 2015 would show acceptable (though not great) performance for young adults initially. Annual reviews in 2016, 2017, 2018 would show aggregate metrics holding steady, providing false reassurance. By 2022, the model would have silently transitioned from "suboptimal but usable" to "completely failed" for young populations—all while administrators saw green lights on their dashboards.

5.3. Race-Based Fairness Stability

Race tells a completely different story. All four race groups showed F1 changes under 1pp. White went from 41.3% to 41.5% (+0.2pp). Black dropped slightly from 50.3% to 49.6% (-0.6pp). Hispanic improved from 48.1% to 48.6% (+0.5pp). More importantly, minority groups outperformed the White population in both years. Black populations achieved

Table 3: Intersectional Performance (Age × Race) - Selected Groups with 95% Bootstrap CIs

Age	Race	N 2022	Diabetes %	F1 2015 [95% CI]	F1 2022 [95% CI]	Change
18-24	White	15,341	1.69%	7.0% [2.8, 11.7]	1.4% [0.0, 3.6]	-5.6pp
18-24	Black	2,311	2.29%	27.0% [6.7, 46.5]	3.3% [0.0, 10.7]	**-23.7pp**
18-24	Hispanic	5,130	2.16%	16.4% [4.3, 30.1]	12.6% [4.9, 21.0]	-3.8pp
80+	White	31,080	18.68%	37.8% [37.0, 38.6]	38.5% [37.7, 39.3]	+0.7pp
80+	Black	1,924	34.25%	50.9% [48.6, 53.4]	51.9% [49.5, 54.2]	+1.0pp
80+	Hispanic	1,103	35.00%	50.9% [47.5, 54.2]	53.9% [50.5, 56.8]	+3.0pp

95% confidence intervals computed via 1,000 bootstrap resamples. Young Black adults experienced worst degradation of any demographic group (-23.7pp); wide CIs reflect small subgroup sizes (53 diabetics in 2022), underscoring the fragility of predictions for low-prevalence intersections. Elderly CIs are tight (±1-3pp), confirming stable performance. Complete 52-intersection matrix in Supplementary Table S5.

49.6% F1 in 2022 compared to White's 41.5%—an 8.1pp advantage. This stems from baseline prevalence: Black populations had 21.44% diabetes prevalence versus White's 14.13%, providing stronger training signal (complete race-stratified analysis in Supplementary Section S5). However, this aggregate stability masks intersectional harm.

5.4. Intersectional Analysis: Young Black Adults

When we break down performance by age and race simultaneously, a different pattern emerges. Table 3 shows F1-scores for key age×race combinations. Young Black adults suffered the worst degradation of any group: F1-score collapsed from 27.0% in 2015 to 3.3% in 2022—a 23.7pp drop representing 88% performance loss. Recall fell from 15.6% to 1.9%. Out of 53 young Black diabetics in the 2022 test set, the model identified 1 and missed 52.

This contradicts the race-only analysis where overall Black performance stayed stable at ∼50% F1. Young Black adults degraded 4.2 times worse than young White adults (-23.7pp vs. -5.6pp). Young Black adults had 1.49% baseline prevalence increasing to 2.29%—a 54% relative increase on a tiny training sample of 32 diabetics. The model learned "young + Black = very low risk" from 2015 and could not update. Feature importance (SHAP) analysis shows BMI plummeted from top-3 to 9th position for this subgroup, indicating complete feature-outcome decoupling (detailed intersectional mechanism in Supplementary Section S11). Figure 5 shows the full age×race landscape.

A standard fairness audit examining race and age separately would conclude: (1) young adults suffer (true), (2) Black adults are protected (true in aggregate), therefore (3) young Black adults are likely okay (catastrophically false). Only intersectional analysis reveals the 88% performance collapse.

5.5. COVID-19 as Distribution Shift

The 2020 data provides a natural experiment in how external shocks create distribution shifts. Diabetes prevalence plateaued at 14.20% in 2020 (expected: 14.91% from trend), then jumped to 15.02% by 2022. The 2020 dip is visible across all demographic groups. This isn't real health improvement—it's diagnostic suppression. The COVID-19 pandemic

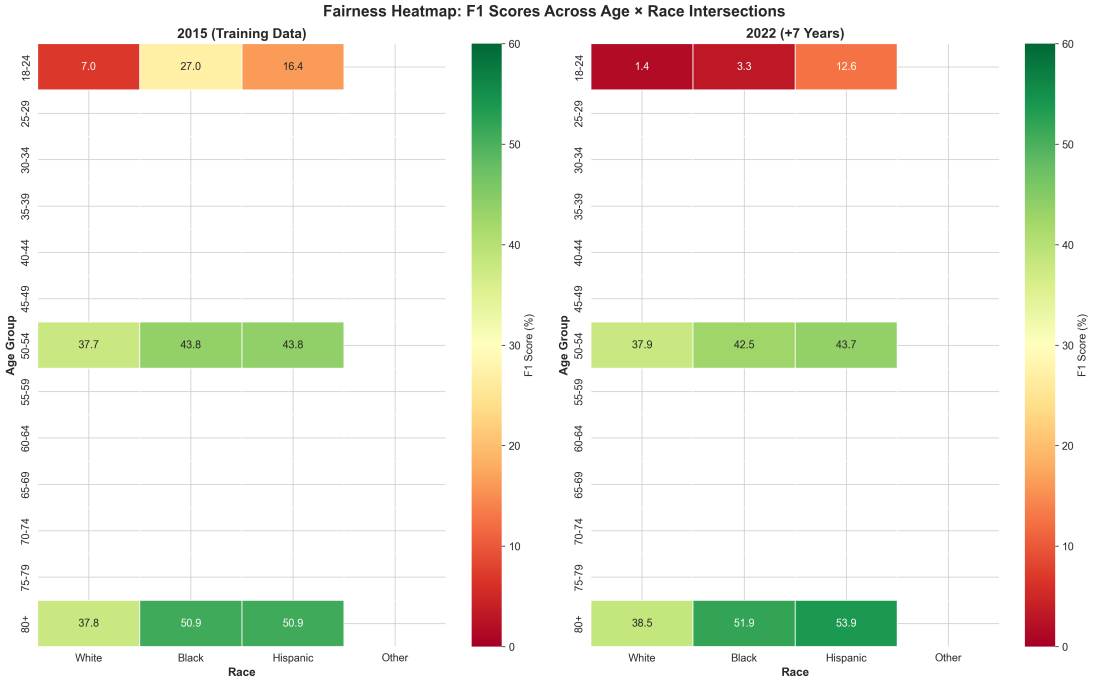

Figure 5: Fairness heatmap across all age×race intersections. Left panel shows 2015 F1-scores. Right panel shows 2022. Red indicates poor performance (<20% F1), green indicates good (>40%). Young rows (18-34) turned deep red by 2022, especially for Black and Hispanic columns. Older rows (65+) remained green across all races.

forced millions to delay routine checkups, temporarily decoupling risk factors from reported diagnosis. By 2022, healthcare access normalized and the feature-outcome relationships restored. This demonstrates why retraining on temporarily corrupted data introduces new biases (complete COVID-19 analysis in Supplementary Section S14).

6. Discussion

6.1. The Stability Paradox in Deployed Healthcare AI

Our findings reveal that the "stability" reported in many healthcare AI deployments is often a statistical illusion. While our model maintained a consistent 71-72% accuracy and 43% F1-score across seven years—metrics that would satisfy most standard deployment audits—these aggregate numbers acted as a veil. The model experienced catastrophic collapse for young adults, particularly young Black adults who suffered an 88% performance loss. This paradox occurs because the statistical gains in the majority older population effectively "cancel out" the abandonment of the minority younger population in the final averaged score.

The clinical implications are severe. Healthcare systems typically review deployed AI models annually or semi-annually, checking aggregate performance dashboards. Our findings show this approach fails. A model can maintain 71% accuracy while systemat-

ically missing 98% of young diabetics. The solution requires transitioning from macro-performance tracking to micro-equity monitoring: computing stratified metrics for every protected group and intersection at every evaluation cycle.

6.2. Mechanisms of Intersectional Algorithmic Harm

Young Black adults experience 4.2 times worse degradation than young White adults through three compounding factors. First, low baseline prevalence (1.49%) creates fragile predictions with only 32 training examples. Second, rapid relative prevalence change (54% increase) breaks static decision boundaries. Third, intersectional invisibility means separate age and race audits miss the -23.7pp degradation. This reveals a critical gap: most fairness implementations examine protected attributes in isolation because intersectionality requires monitoring 52 intersections instead of 17 groups. However, our results prove that "checking the boxes" independently is not just insufficient; it is misleading (detailed mechanism in Supplementary Section S11).

The policy implication is clear: fairness audits for healthcare AI must be intersectional by default, not as an optional enhancement. Regulatory frameworks and institutional review boards should require stratified performance reporting for age×race, age×sex, and race×sex at minimum, following TRIPOD+AI Collins et al. (2024) and fairness policy recommendations Pfohl et al. (2022).

6.3. Implementation: Three-Tier Monitoring Framework

Healthcare organizations should implement tiered fairness monitoring balancing cost and safety. **Tier 1 (Quarterly):** Track aggregate accuracy, F1, AUC-ROC; escalate if any drops >3pp. **Tier 2 (Bi-annual):** Compute stratified metrics for age, race, sex, income separately; escalate if any group shows >5pp degradation. **Tier 3 (Annual):** Compute full age×race×sex performance; trigger mandatory retraining if any intersection shows >10pp degradation or drops below 20% F1. The escalation structure balances cost and safety: most quarters pass Tier 1, Tier 2 catches gradual drift, Tier 3 catches rare but catastrophic intersectional failures (complete framework in Supplementary Section S12).

6.4. Retraining as Mitigation

To move beyond speculation, we retrained XGBoost on combined 2015+2020 data (N=812,398) and evaluated on 2022. Overall performance was largely unchanged (F1: 43.4% → 43.2%), but intersectional results were mixed. Young Hispanic adults recovered nearly all lost performance (F1: 12.6% → 16.2%, 94% recovery), while young White adults recovered only 25% (1.4% → 2.8%). Young Black adults—the worst-affected group—improved from 3.3% to 5.9% F1, recovering just 11% of the 23.7pp loss. Elderly subgroups were unaffected (±0.3pp). These results demonstrate that retraining on more recent data provides marginal relief for the most vulnerable intersections but is fundamentally insufficient as a standalone mitigation. The 2020 data also carries COVID-era diagnostic suppression, risking the introduction of new biases (Supplementary Section S14). We recommend that retraining be paired with intersectional monitoring: triggered only when Tier 3 audits flag >10pp degradation and validated against subgroup-level benchmarks before deployment.

6.5. Generalizability Beyond Diabetes

The stability paradox likely generalizes to other healthcare prediction tasks. Any domain with (1) strong age gradients in prevalence, (2) low-prevalence subgroups, and (3) shifting population distributions will exhibit similar dynamics. Cardiovascular disease prediction, cancer screening, and mental health risk assessment all share these properties. Beyond healthcare, criminal justice risk assessment, loan default prediction, and college admissions face similar risks. Organizations deploying AI should ask: Which subgroups have low prevalence (<5%)? How is prevalence changing? If prevalence increases rapidly (>20% relative change) for a low-baseline group, that group is at high risk of catastrophic fairness degradation.

7. Limitations

While our longitudinal analysis provides a rare, large-scale view of temporal fairness, several constraints must be acknowledged. First, we rely on self-reported health data subject to recall bias. The "diagnostic suppression" in 2020 reflects that many individuals could not access healthcare for diagnosis during COVID-19, meaning our labels are proxies for "clinical interaction." Second, race harmonization required aggregating Asian and Pacific Islander respondents into a single category to maintain 2015 compatibility, masking unique health trajectories within these communities. Third, while our primary analysis uses XGBoost to simulate typical "set-and-forget" deployment, we additionally evaluated logistic regression and a retraining strategy to test generalizability; bootstrap confidence intervals (1,000 resamples) quantify metric uncertainty for small subgroups, though the wide CIs for young Black adults ([0.0, 10.7]% in 2022) underscore the inherent instability of predictions at these sample sizes. Finally, while BRFSS is massive and geographically diverse, it is limited to the United States; temporal dynamics may differ globally. Our models were trained and evaluated without incorporating BRFSS survey weights. While unweighted analysis may yield prevalence estimates that differ from nationally representative figures, our primary objective measuring *relative* performance changes across demographic subgroups over time is less sensitive to absolute prevalence calibration than population-level inference would be. Nonetheless, differential non-response patterns across demographic groups could affect subgroup-level metric comparisons, and future work should explore survey-weight-aware modeling approaches.

8. Conclusion

We demonstrate that machine learning models for healthcare can maintain stable aggregate performance while experiencing catastrophic fairness degradation for vulnerable subgroups—a phenomenon we term the stability paradox. Training an XGBoost diabetes prediction model on 2015 BRFSS data and testing on 2022 data, we found overall accuracy remained at 71% and F1-scores held steady at 43% across seven years. Yet young adults aged 18-29 experienced severe performance collapse, with recall falling from 6-12% to 2-7%. Critically, intersectional analysis revealed that young Black adults suffered the worst degradation: F1-scores collapsed from 27.0% to 3.3%, representing an 88% performance loss.

This finding contradicts aggregate race-based fairness metrics, which showed Black populations outperforming White populations by 8.1pp. Standard fairness audits examining age and race separately would completely miss the catastrophic failure concentrated at the young×Black intersection. We identify the mechanism: groups with low baseline prevalence (<2%) cannot adapt to rapid relative prevalence changes (40-54% increases) even when absolute changes are small.

Our findings establish four imperatives for responsible healthcare AI deployment. First, temporal fairness monitoring must transition from macro-performance tracking to intersectional micro-equity auditing. Second, fairness audits must be intersectional by default. Third, model architecture matters: nonlinear models like XGBoost amplify temporal fairness degradation by 7.5× relative to Logistic Regression for the most vulnerable subgroups, and deployment decisions should weigh this trade-off against marginal aggregate performance gains. Fourth, retraining alone is insufficient, our experiment recovering only 11% of lost performance for young Black adults demonstrates that organizations must pair retraining with intersectional monitoring and subgroup-level validation. The stability paradox is not unique to diabetes prediction, any domain with demographic gradients in outcome prevalence, low-prevalence subgroups, and shifting populations will exhibit similar dynamics.

Future work should extend this analysis to other health outcomes, test whether different model architectures exhibit similar patterns, develop algorithmic approaches for fairness-aware continual learning, and investigate mitigation strategies beyond periodic retraining. The AI research community often treats fairness as a deployment-time concern: ensure the model is fair when launched, then monitor for performance decay. Our findings show this approach fails. Fairness is not a fixed property but a temporal process. Models that are fair today can become deeply unfair tomorrow, and the degradation concentrates on precisely the populations most vulnerable to healthcare inequities.

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